

Dionex_COMPASS_May_2023_rerun_a

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Daily Set up

Read in and Format the raw data - change wd & file names

```
# SULFATE DATA:
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Sdat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202305_rerun_S04a.txt", sep='\t' , header=T, skip=1)  
head(Sdat)
```

```
##      X          X.1          X.2 IC.S04 IC.S04.1 IC.S04.2 IC.S04.3 IC.S04.4  
## 1 1 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 2 2 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 3 3 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 4 4 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 5 5 Standard 1 Calibration Standard 4.18 0.4677 6.5 0.423 3.54  
## 6 6 Standard 2 Calibration Standard 4.18 0.9552 6.8 0.8641 7.13
```

```
## Only keep the columns that we need
```

```
Sdat <- Sdat[,c(2,5)] # dont need this here  
head(Sdat)
```

```
##      X.1 IC.S04.1  
## 1 Lab Blank    n.a.  
## 2 Lab Blank    n.a.  
## 3 Lab Blank    n.a.  
## 4 Lab Blank    n.a.  
## 5 Standard 1 0.4677  
## 6 Standard 2 0.9552
```

```
## Name the columns correctly
```

```
colnames(Sdat) <- c("Sample_ID", "S04_ppm")  
Sdat$Sample_ID <- as.factor(Sdat$Sample_ID)  
Sdat$S04_ppm <- as.numeric(Sdat$S04_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat <- as.data.frame(Sdat)
head(Sdat)
```

```
##      Sample_ID S04_ppm
## 1 Lab Blank      NA
## 2 Lab Blank      NA
## 3 Lab Blank      NA
## 4 Lab Blank      NA
## 5 Standard 1  0.4677
## 6 Standard 2  0.9552
```

```
#Chloride data
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Cldat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202305_rerun_Cla.txt",sep='\t' , header=T, skip=1)
head(Cldat)
```

```
##      X      X.1      X.2 IC.Cl IC.Cl.1 IC.Cl.2 IC.Cl.3 IC.Cl.4
## 1 1 Lab Blank      Unknown n.a.      n.a.      n.a.      n.a.      n.a.
## 2 2 Lab Blank      Unknown n.a.      n.a.      n.a.      n.a.      n.a.
## 3 3 Lab Blank      Unknown n.a.      n.a.      n.a.      n.a.      n.a.
## 4 4 Lab Blank      Unknown n.a.      n.a.      n.a.      n.a.      n.a.
## 5 5 Standard 1 Calibration Standard 3.083 5.0741 92.54 6.0218 62.99
## 6 6 Standard 2 Calibration Standard 3.083 9.9643 93.11 11.8254 122.49
```

```
## Only keep the columns that we need
```

```
Cldat <- Cldat[,c(2,5)]
head(Cldat)
```

```
##      X.1 IC.Cl.1
## 1 Lab Blank  n.a.
## 2 Lab Blank  n.a.
## 3 Lab Blank  n.a.
## 4 Lab Blank  n.a.
## 5 Standard 1 5.0741
## 6 Standard 2 9.9643
```

```
## Name the columns correctly
```

```
colnames(Cldat) <- c("Sample_ID", "Cl_ppm")
Cldat$Sample_ID <- as.factor(Cldat$Sample_ID)
Cldat$Cl_ppm <- as.numeric(Cldat$Cl_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat <- as.data.frame(Cldat)
head(Cldat)
```

```
##      Sample_ID Cl_ppm
## 1 Lab Blank      NA
## 2 Lab Blank      NA
## 3 Lab Blank      NA
## 4 Lab Blank      NA
## 5 Standard 1 5.0741
## 6 Standard 2 9.9643
```

```
## Bring the data back together:
all_dat <- merge(Sdat, Cldat, by="Sample_ID")
head(all_dat)
```

```
##      Sample_ID  S04_ppm    Cl_ppm
## 1           1890.423 11757.423
## 2           1890.423  1864.499
## 3           1890.423    0.011
## 4           1890.423  3685.568
## 5           1890.423      NA
## 6           270.227 11757.423
```

```
## Remove empty lines
all_dat <- all_dat[!(is.na(all_dat$Sample_ID) | all_dat$Sample_ID==""), ]
head(all_dat)
```

```
##              Sample_ID  S04_ppm    Cl_ppm
## 26      100_GWI_202305_WC_LysB_20cm 1227.5264  9239.344
## 27 100_GWI_202305_WC_LysB_20cm_dup 1205.4744  9003.641
## 28      101_GWI_202305_WC_LysB_45cm 1376.7938  8628.659
## 29 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.510
## 30      102_GWI_202305_WC_LysC_10cm  958.8118  7841.280
## 31      103_GWI_202305_WC_LysC_20cm 1411.9603  9624.554
```

```
all_dat <- droplevels(all_dat[!all_dat$Sample_ID == 'Lab Blank',])
head(all_dat)
```

```
##              Sample_ID  S04_ppm    Cl_ppm
## 26      100_GWI_202305_WC_LysB_20cm 1227.5264  9239.344
## 27 100_GWI_202305_WC_LysB_20cm_dup 1205.4744  9003.641
## 28      101_GWI_202305_WC_LysB_45cm 1376.7938  8628.659
## 29 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.510
## 30      102_GWI_202305_WC_LysC_10cm  958.8118  7841.280
## 31      103_GWI_202305_WC_LysC_20cm 1411.9603  9624.554
```

#Note from Steph: Check all sample IDs from MSM Upland in 2023 May samples. Samples were labeled LysA, but are actually LysC from this month and zone only and need documented. If there are samples from MSM UP LysA 10cm listed, then the samples were NOT corrected and that would mean that all of the LysA samples should be LysC and we should document and correct. We know this because we hardly ever get water from LysA 10cm at MSM. This is a one time issue from May 2023 as far as I am aware.

```
## [1] "The string 'MSM_202305_UP_LysA_10cm' was not found in the dataframe."
```

Pull out standards - could do some checks here if we want

```
stds <- all_dat[grep("Standard", all_dat$Sample_ID),]
head(stds)
```

```
##      Sample_ID S04_ppm Cl_ppm
## 208 Standard 1  0.4677 5.0741
## 209 Standard 1  0.4677 5.1730
## 210 Standard 1  0.4677 5.1059
## 211 Standard 1  0.4723 5.0741
## 212 Standard 1  0.4723 5.1730
## 213 Standard 1  0.4723 5.1059
```

```
stds_chk_S <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(S04_ppm), sd = sd(S04_ppm))
stds_chk_S$cv <- (stds_chk_S$sd/stds_chk_S$mean)*100
stds_chk_S$flag <- ifelse(stds_chk_S$cv <2, 'YES', 'NO, rerun')
head(stds_chk_S)
```

```
## # A tibble: 5 x 5
##   Sample_ID   mean      sd      cv flag
##   <fct>      <dbl>   <dbl> <dbl> <chr>
## 1 Standard 1  0.470 0.00207 0.440 YES
## 2 Standard 2  0.956 0.000910 0.0951 YES
## 3 Standard 3  1.95 0.00337 0.173 YES
## 4 Standard 4 10.0 0.00104 0.0104 YES
## 5 Standard 5 20.0 0.00346 0.0173 YES
```

```
stds_chk_Cl <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(Cl_ppm), sd = sd(Cl_ppm))
stds_chk_Cl$cv <- (stds_chk_Cl$sd/stds_chk_Cl$mean)*100
stds_chk_Cl$flag <- ifelse(stds_chk_Cl$cv <2, 'YES', 'NO, rerun')
head(stds_chk_Cl)
```

```
## # A tibble: 5 x 5
##   Sample_ID   mean      sd      cv flag
##   <fct>      <dbl>   <dbl> <dbl> <chr>
## 1 Standard 1  5.12 0.0437 0.854 YES
## 2 Standard 2  9.95 0.0645 0.649 YES
## 3 Standard 3 19.9 0.0522 0.262 YES
## 4 Standard 4 100. 0.0274 0.0273 YES
## 5 Standard 5 200. 0.0839 0.0420 YES
```

Calculate mmol/L concentrations

```
#remove standards from sample dataframe
sampledat <- all_dat[!grepl("Standard", all_dat$Sample_ID),]
head(sampledat)
```

```
##           Sample_ID S04_ppm Cl_ppm
## 26 100_GWI_202305_WC_LysB_20cm 1227.5264 9239.344
## 27 100_GWI_202305_WC_LysB_20cm_dup 1205.4744 9003.641
## 28 101_GWI_202305_WC_LysB_45cm 1376.7938 8628.659
```

```
## 29 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.510
## 30      102_GWI_202305_WC_LysC_10cm  958.8118  7841.280
## 31      103_GWI_202305_WC_LysC_20cm 1411.9603  9624.554
```

```
# Constants needed for calculations:
```

```
clmw <- 35.45      #molecular weight of Chloride: 35.45
```

```
smw <- 32.06      #molecular weight of sulfur: 32.06
```

```
# Convert ppm to mmol/L
```

```
sampledat$S04_mM <- (sampledat$S04_ppm / smw)
```

```
sampledat$Cl_mM <- (sampledat$Cl_ppm / clmw)
```

```
# Calculate Salinity
```

```
# calculated using the Knudsen equation
```

```
# Salinity = 0.03 + 1.8050 * Chlorinity
```

```
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
```

```
# =((1.807*Cl_ppm)+0.026)/1000
```

```
sampledat$salinity <- ((1.8070 * sampledat$Cl_ppm) + 0.026) / 1000
```

```
head(sampledat)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 26      100_GWI_202305_WC_LysB_20cm 1227.5264  9239.344 38.28841 260.6303
## 27 100_GWI_202305_WC_LysB_20cm_dup 1205.4744  9003.641 37.60057 253.9814
## 28      101_GWI_202305_WC_LysB_45cm 1376.7938  8628.659 42.94429 243.4037
## 29 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.510 58.96516 324.4996
## 30      102_GWI_202305_WC_LysC_10cm  958.8118  7841.280 29.90679 221.1927
## 31      103_GWI_202305_WC_LysC_20cm 1411.9603  9624.554 44.04118 271.4966
##      salinity
## 26 16.69552
## 27 16.26960
## 28 15.59201
## 29 20.78687
## 30 14.16922
## 31 17.39159
```

Pull out dups and check with percent difference

```
#Show me the data that we have from the calculations
```

```
head(sampledat)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 26      100_GWI_202305_WC_LysB_20cm 1227.5264  9239.344 38.28841 260.6303
## 27 100_GWI_202305_WC_LysB_20cm_dup 1205.4744  9003.641 37.60057 253.9814
## 28      101_GWI_202305_WC_LysB_45cm 1376.7938  8628.659 42.94429 243.4037
## 29 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.510 58.96516 324.4996
## 30      102_GWI_202305_WC_LysC_10cm  958.8118  7841.280 29.90679 221.1927
## 31      103_GWI_202305_WC_LysC_20cm 1411.9603  9624.554 44.04118 271.4966
```

```
##      salinity
## 26 16.69552
## 27 16.26960
## 28 15.59201
## 29 20.78687
## 30 14.16922
## 31 17.39159
```

```
#pull out any rows that have "dup" in the SampleID column
dups <- sampledats %>%
  filter(str_detect(Sample_ID, "dup"))      #have to change this to match data
head(dups)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM
## 1 100_GWI_202305_WC_LysB_20cm_dup 1205.4744 9003.6408 37.60057392
## 2 110_SWH_202305_UP_LysB_10cm_dup   0.3371  35.9107  0.01051466
## 3 120_SWH_202305_UPCON_LysB_20cm_dup 16.1193   5.2771  0.50278540
## 4 130_SWH_202305_TR_LysB_45cm_dup   0.6169  426.1415  0.01924205
## 5 140_SWH_202305_WC_LysC_10cm_dup   1.0043  208.4391  0.03132564
##              Cl_mM      salinity
## 1 253.9814048 16.26960493
## 2  1.0129958  0.06491663
## 3  0.1488604  0.00956172
## 4 12.0209168  0.77006369
## 5  5.8798054  0.37667545
```

```
#remove these from sample dataframe in a new dataframe
sampledat2 <- sampledats %>%
  filter(!str_detect(Sample_ID, "dup")) %>%
  filter(!str_detect(Sample_ID, "spk"))
head(sampledat2)
```

```
##              Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM salinity
## 1 100_GWI_202305_WC_LysB_20cm 1227.5264 9239.344 38.28841 260.6303 16.69552
## 2 101_GWI_202305_WC_LysB_45cm 1376.7938 8628.659 42.94429 243.4037 15.59201
## 3 102_GWI_202305_WC_LysC_10cm  958.8118 7841.280 29.90679 221.1927 14.16922
## 4 103_GWI_202305_WC_LysC_20cm 1411.9603 9624.554 44.04118 271.4966 17.39159
## 5      104_GWI_202305_SW_A 1574.3848 11468.676 49.10745 323.5170 20.72392
## 6      105_GWI_202305_SW_B 1609.8636 11757.423 50.21409 331.6621 21.24569
```

```
#remove the dup from these IDs so we will have duplicate sample names
dups$Sample_ID<-gsub("_dup", "", as.character(dups$Sample_ID))
dups <- dups[, -c(2,3)]
colnames(dups) <- c('Sample_ID', 'S04_mM_dup', 'Cl_mM_dup', 'salinity_dup')
head(dups)
```

```
##              Sample_ID  S04_mM_dup  Cl_mM_dup  salinity_dup
## 1 100_GWI_202305_WC_LysB_20cm 37.60057392 253.9814048 16.26960493
## 2 110_SWH_202305_UP_LysB_10cm 0.01051466  1.0129958  0.06491663
## 3 120_SWH_202305_UPCON_LysB_20cm 0.50278540  0.1488604  0.00956172
## 4 130_SWH_202305_TR_LysB_45cm 0.01924205 12.0209168  0.77006369
## 5 140_SWH_202305_WC_LysC_10cm 0.03132564  5.8798054  0.37667545
```

```
#put it back together with the old data set and look for duplicates
```

```
QAdups <- merge(sampldat2, dups)
```

```
head(QAdups)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1  100_GWI_202305_WC_LysB_20cm 1227.5264 9239.3442 38.28840923 260.6303018
## 2   110_SWH_202305_UP_LysB_10cm   0.3694  39.5784  0.01152215  1.1164570
## 3 120_SWH_202305_UPCON_LysB_20cm  15.8897   5.2152  0.49562383  0.1471142
## 4   130_SWH_202305_TR_LysB_45cm   0.6961 460.6917  0.02171241 12.9955346
## 5   140_SWH_202305_WC_LysC_10cm   0.9558 191.2575  0.02981285  5.3951340
##      salinity  S04_mM_dup  Cl_mM_dup salinity_dup
## 1 16.695520969 37.60057392 253.9814048 16.26960493
## 2  0.071544169  0.01051466  1.0129958  0.06491663
## 3  0.009449866  0.50278540  0.1488604  0.00956172
## 4  0.832495902  0.01924205 12.0209168  0.77006369
## 5  0.345628302  0.03132564  5.8798054  0.37667545
```

```
QAdups$S04_dups_chk <- ((abs(QAdups$S04_mM-QAdups$S04_mM_dup))/((QAdups$S04_mM+QAdups$S04_mM_dup)/2))*100
```

```
QAdups$S04_dups_flag <- ifelse(QAdups$S04_dups_chk <10, 'YES', 'NO, rerun')
```

```
QAdups$Cl_dups_chk <- ((abs(QAdups$Cl_mM-QAdups$Cl_mM_dup))/((QAdups$Cl_mM+QAdups$Cl_mM_dup)/2))*100
```

```
QAdups$Cl_dups_flag <- ifelse(QAdups$Cl_dups_chk <10, 'YES', 'NO, rerun')
```

```
head(QAdups)
```

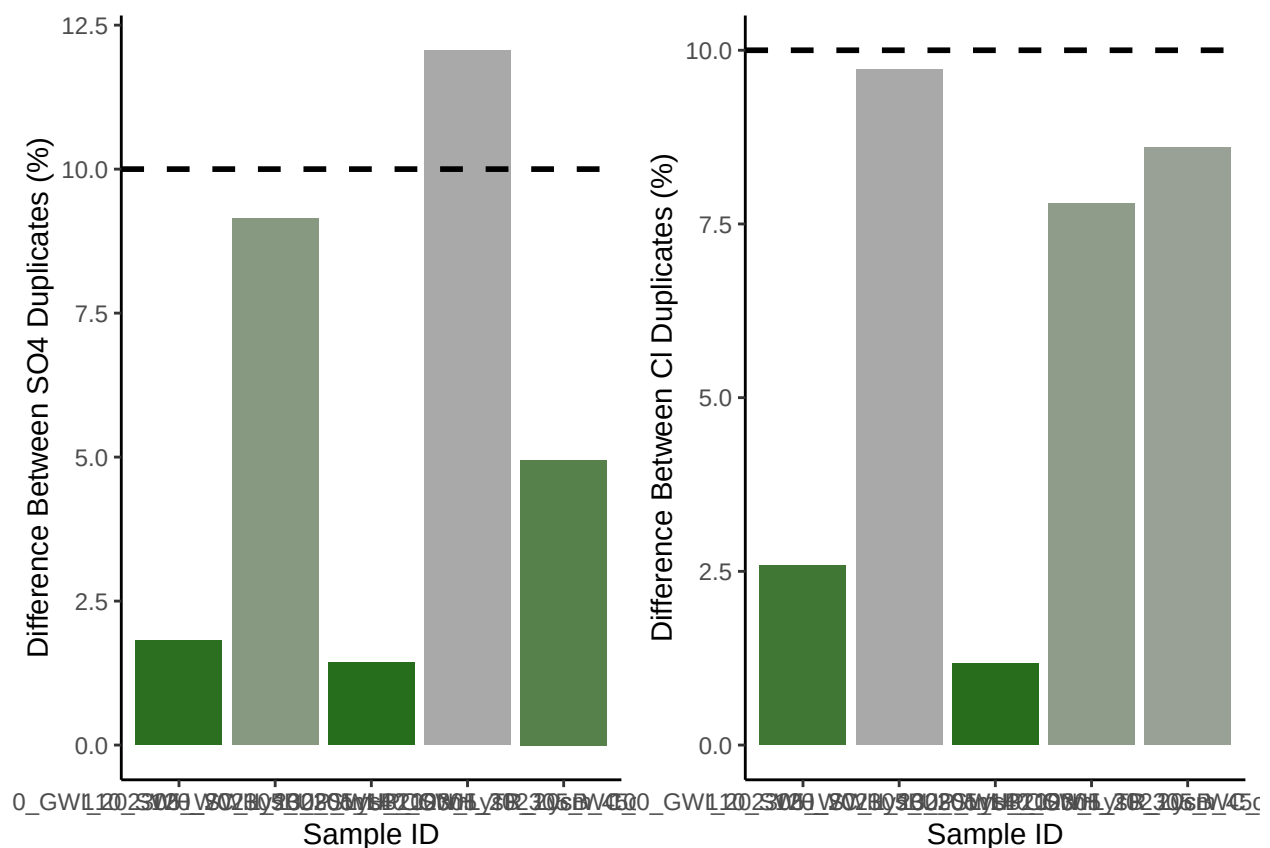
```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1  100_GWI_202305_WC_LysB_20cm 1227.5264 9239.3442 38.28840923 260.6303018
## 2   110_SWH_202305_UP_LysB_10cm   0.3694  39.5784  0.01152215  1.1164570
## 3 120_SWH_202305_UPCON_LysB_20cm  15.8897   5.2152  0.49562383  0.1471142
## 4   130_SWH_202305_TR_LysB_45cm   0.6961 460.6917  0.02171241 12.9955346
## 5   140_SWH_202305_WC_LysC_10cm   0.9558 191.2575  0.02981285  5.3951340
##      salinity  S04_mM_dup  Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 16.695520969 37.60057392 253.9814048 16.26960493      1.812741      YES
## 2  0.071544169  0.01051466  1.0129958  0.06491663      9.143666      YES
## 3  0.009449866  0.50278540  0.1488604  0.00956172      1.434597      YES
## 4  0.832495902  0.01924205 12.0209168  0.77006369     12.063976  NO, rerun
## 5  0.345628302  0.03132564  5.8798054  0.37667545      4.948727      YES
##      Cl_dups_chk Cl_dups_flag
## 1      2.584044      YES
## 2      9.717164      YES
## 3      1.179913      YES
## 4      7.791815      YES
## 5      8.597321      YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
```

```
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_chk, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```
Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)
```



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```
Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##      Flag S_no_rows Flag Cl_no_rows Total S_Percent Cl_Percent
## 1 NO, rerun      1 YES           5      5         20        100
## 2      YES      4 YES           5      5         80        100
```

Pull out dups and check with cv

```
#the cv
# calculate the sd bewteen the two duplicates then divide by the mean and multiply by 100

df2 <- as.data.frame(QAdups$S04_mM)
df2$dups <- QAdups$S04_mM_dup

df2$sds <- apply(df2,1,sd)

QAdups$S04_dups_cv <- (df2$sds)/((QAdups$S04_mM+QAdups$S04_mM_dup)/2) * 100
QAdups$S04_dups_cv_flag <- ifelse(QAdups$S04_dups_cv <11, 'YES', 'NO, rerun')

head(QAdups)
```

```
##      Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 100_GWI_202305_WC_LysB_20cm 1227.5264 9239.3442 38.28840923 260.6303018
## 2 110_SWH_202305_UP_LysB_10cm  0.3694  39.5784  0.01152215  1.1164570
## 3 120_SWH_202305_UPCON_LysB_20cm 15.8897   5.2152  0.49562383  0.1471142
## 4 130_SWH_202305_TR_LysB_45cm  0.6961 460.6917  0.02171241 12.9955346
## 5 140_SWH_202305_WC_LysC_10cm  0.9558 191.2575  0.02981285  5.3951340
##      salinity S04_mM_dup  Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 16.695520969 37.60057392 253.9814048 16.26960493 1.812741      YES
## 2 0.071544169 0.01051466 1.0129958 0.06491663 9.143666      YES
## 3 0.009449866 0.50278540 0.1488604 0.00956172 1.434597      YES
## 4 0.832495902 0.01924205 12.0209168 0.77006369 12.063976 NO, rerun
## 5 0.345628302 0.03132564 5.8798054 0.37667545 4.948727      YES
##      Cl_dups_chk Cl_dups_flag S04_dups_cv S04_dups_cv_flag
## 1 2.584044      YES 1.281801      YES
## 2 9.717164      YES 6.465548      YES
## 3 1.179913      YES 1.014413      YES
## 4 7.791815      YES 8.530519      YES
## 5 8.597321      YES 3.499278      YES
```

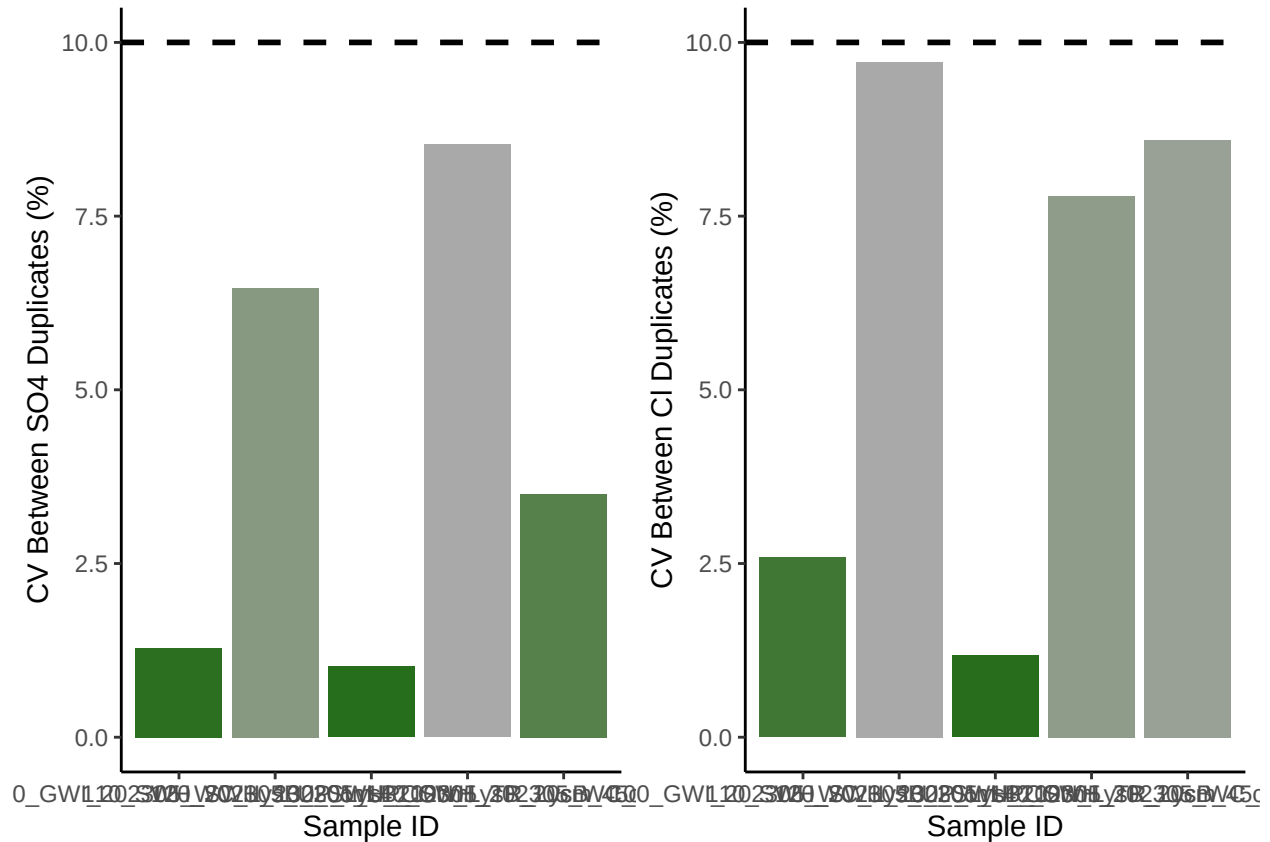
```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_cv, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```

Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
    color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)

```



```

#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(SO4_dups_flag) %>%
  summarise(S_no_rows = length(SO4_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$SO4_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)

```

```
##          Flag S_no_rows Flag Cl_no_rows Total S_Percent Cl_Percent
## 1 NO, rerun      1 YES          5      5         20         100
## 2      YES      4 YES          5      5         80         100
```

Pull out spikes and check

```
#Show me the data that we have from the calculations
head(sampledat)
```

```
##          Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 26 100_GWI_202305_WC_LysB_20cm 1227.5264 9239.344 38.28841 260.6303
## 27 100_GWI_202305_WC_LysB_20cm_dup 1205.4744 9003.641 37.60057 253.9814
## 28 101_GWI_202305_WC_LysB_45cm 1376.7938 8628.659 42.94429 243.4037
## 29 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.510 58.96516 324.4996
## 30 102_GWI_202305_WC_LysC_10cm 958.8118 7841.280 29.90679 221.1927
## 31 103_GWI_202305_WC_LysC_20cm 1411.9603 9624.554 44.04118 271.4966
##      salinity
## 26 16.69552
## 27 16.26960
## 28 15.59201
## 29 20.78687
## 30 14.16922
## 31 17.39159
```

```
#pull out any rows that have "spk" in the SampleID column
spks <- sampledat %>%
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
head(spks)
```

```
##          Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1 101_GWI_202305_WC_LysB_45cm_spk 1890.4229 11503.5095 58.965156 324.4995628
## 2 111_SWH_202305_UP_LysB_20cm_spk 87.0810 27.5305 2.716188 0.7766008
## 3 121_SWH_202305_UPCON_LysB_45cm_spk 117.3079 10.4805 3.659011 0.2956417
## 4 131_SWH_202305_TR_LysC_10cm_spk 130.4766 379.3708 4.069763 10.7015740
## 5 141_SWH_202305_WC_LysC_20cm_spk 85.7365 402.1936 2.674251 11.3453766
##      salinity
## 1 20.78686767
## 2 0.04977361
## 3 0.01896426
## 4 0.68554904
## 5 0.72678984
```

```
#remove the dup from these IDs so we will have duplicate sample names
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
spks <- spks[ , -c(2,3, 5,6)]
colnames(spks) <- c('Sample_ID', 'S04_mM_spk')
head(spks)
```

```
##          Sample_ID S04_mM_spk
## 1 101_GWI_202305_WC_LysB_45cm 58.965156
```

```
## 2    111_SWH_202305_UP_LysB_20cm    2.716188
## 3 121_SWH_202305_UPCON_LysB_45cm    3.659011
## 4    131_SWH_202305_TR_LysC_10cm    4.069763
## 5    141_SWH_202305_WC_LysC_20cm    2.674251
```

#put it back together with the old data set and look for duplicates

```
QAspks <- merge(sampledats, spks)
head(QAspks)
```

```
##           Sample_ID    S04_ppm    Cl_ppm    S04_mM    Cl_mM
## 1 101_GWI_202305_WC_LysB_45cm 1376.7938 8628.6595 42.944285714 243.4036530
## 2 111_SWH_202305_UP_LysB_20cm    0.1088    26.3485 0.003393637    0.7432581
## 3 121_SWH_202305_UPCON_LysB_45cm 27.7212    8.2705 0.864666251    0.2333004
## 4 131_SWH_202305_TR_LysC_10cm 41.1769   362.3567 1.284369931   10.2216276
## 5 141_SWH_202305_WC_LysC_20cm    0.4182   375.1654 0.013044292   10.5829450
##      salinity S04_mM_spk
## 1 15.59201372 58.965156
## 2 0.04763774 2.716188
## 3 0.01497079 3.659011
## 4 0.65480456 4.069763
## 5 0.67794988 2.674251
```

#now we need to calculate the spike concentration and calculate the spike recovery

```
spkconc <- (250/smw)      # in mM
spkvol <- 10              # in uL
spkvol <- spkvol/1000000
#spike for these samples was 10uL of the 250mM standard
QAspks$S04_spk_Conc <- (spkconc)*spkvol      # mmoles of S04
head(QAspks)
```

```
##           Sample_ID    S04_ppm    Cl_ppm    S04_mM    Cl_mM
## 1 101_GWI_202305_WC_LysB_45cm 1376.7938 8628.6595 42.944285714 243.4036530
## 2 111_SWH_202305_UP_LysB_20cm    0.1088    26.3485 0.003393637    0.7432581
## 3 121_SWH_202305_UPCON_LysB_45cm 27.7212    8.2705 0.864666251    0.2333004
## 4 131_SWH_202305_TR_LysC_10cm 41.1769   362.3567 1.284369931   10.2216276
## 5 141_SWH_202305_WC_LysC_20cm    0.4182   375.1654 0.013044292   10.5829450
##      salinity S04_mM_spk S04_spk_Conc
## 1 15.59201372 58.965156 7.797879e-05
## 2 0.04763774 2.716188 7.797879e-05
## 3 0.01497079 3.659011 7.797879e-05
## 4 0.65480456 4.069763 7.797879e-05
## 5 0.67794988 2.674251 7.797879e-05
```

#need to determine dilution factors and initial amount of sample added

#if your samples are all the same dilution just use the first line

#for Steph / COMPASS this depends on the site so...

```
QAspks$Dilution <- 1
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "UP"), 50, 1)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "TR"), 50, 1)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "WC"), 100, 1)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "SW"), 100, 1)
```

```

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "UP"), 50,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "TR"), 50,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "WC"), 100,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "SW"), 100,

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "UP"), 100,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "TR"), 100,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "WC"), 200,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "SW"), 200,

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH"), 50, QAspks$Dilution)

#Set Sample volumes in uL
QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(QAspks$Dilution == 50, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 100, 1466, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 200, 1458, QAspks$SampleVol)

#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000000
head(QAspks)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  101_GWI_202305_WC_LysB_45cm 1376.7938 8628.6595 42.944285714 243.4036530
## 2   111_SWH_202305_UP_LysB_20cm   0.1088  26.3485  0.003393637  0.7432581
## 3 121_SWH_202305_UPCON_LysB_45cm  27.7212   8.2705  0.864666251  0.2333004
## 4   131_SWH_202305_TR_LysC_10cm  41.1769  362.3567  1.284369931  10.2216276
## 5   141_SWH_202305_WC_LysC_20cm   0.4182  375.1654  0.013044292  10.5829450
##      salinity S04_mM_spk S04_spk_Conc Dilution SampleVol
## 1 15.59201372  58.965156 7.797879e-05      200  0.001458
## 2  0.04763774   2.716188 7.797879e-05       50  0.001483
## 3  0.01497079   3.659011 7.797879e-05       50  0.001483
## 4  0.65480456   4.069763 7.797879e-05       50  0.001483
## 5  0.67794988   2.674251 7.797879e-05       50  0.001483

```

```

#gives us the total S04 in the sample in mmoles
QAspks$S04_Total_unspkd <- (QAspks$S04_mM/QAspks$Dilution)*(QAspks$SampleVol)

##total S04 in spiked sample in mmoles
QAspks$S04_Total_spkd <- (QAspks$S04_mM_spk/QAspks$Dilution)*(QAspks$SampleVol+spkvol)

QAspks$S04_expctd_spkd <- (QAspks$S04_Total_unspkd + QAspks$S04_spk_Conc)
QAspks$spk_recovery <- (QAspks$S04_Total_spkd/QAspks$S04_expctd_spkd)*100
QAspks$S04_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO', rerun=TRUE)

head(QAspks)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  101_GWI_202305_WC_LysB_45cm 1376.7938 8628.6595 42.944285714 243.4036530
## 2   111_SWH_202305_UP_LysB_20cm   0.1088  26.3485  0.003393637  0.7432581
## 3 121_SWH_202305_UPCON_LysB_45cm  27.7212   8.2705  0.864666251  0.2333004

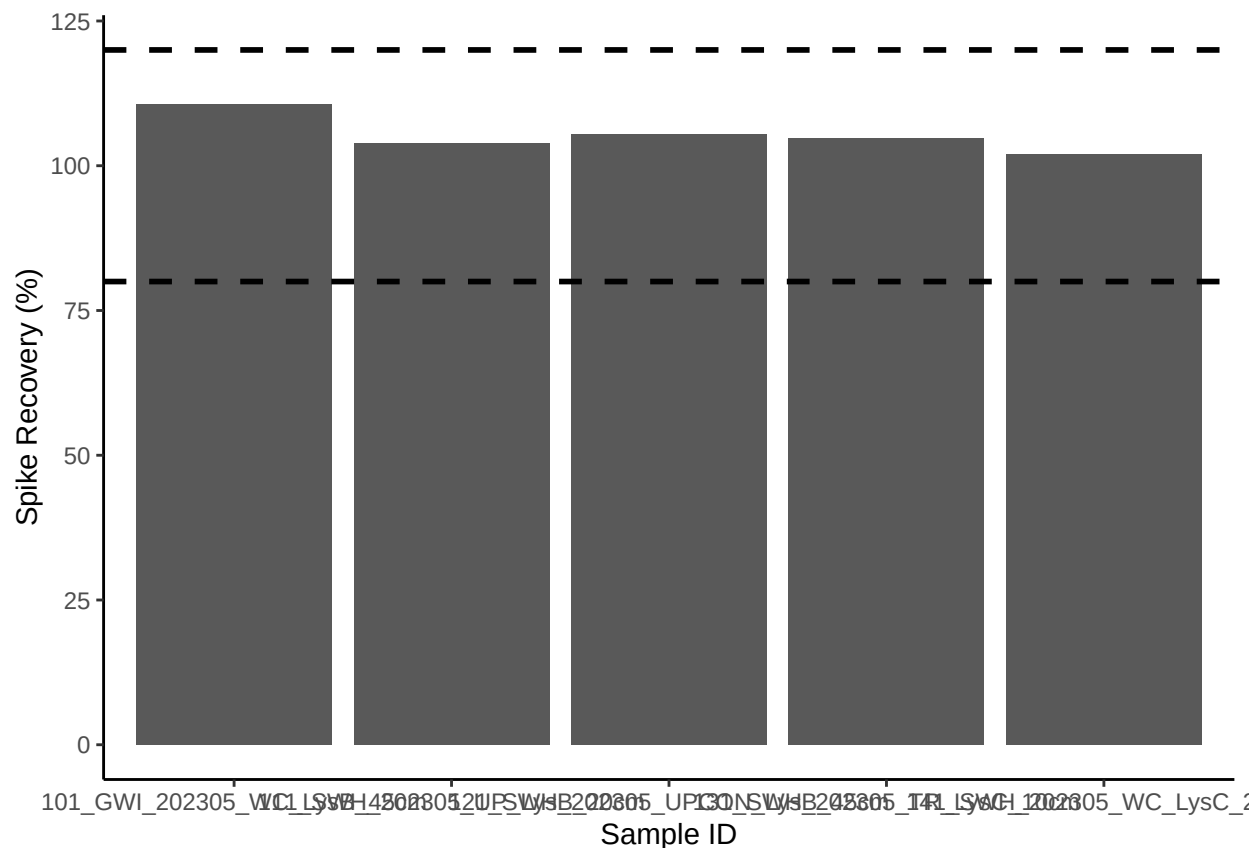
```

```
## 4 131_SWH_202305_TR_LysC_10cm 41.1769 362.3567 1.284369931 10.2216276
## 5 141_SWH_202305_WC_LysC_20cm 0.4182 375.1654 0.013044292 10.5829450
## salinity S04_mM_spk S04_spk_Conc Dilution SampleVol S04_Total_unspkd
## 1 15.59201372 58.965156 7.797879e-05 200 0.001458 3.130638e-04
## 2 0.04763774 2.716188 7.797879e-05 50 0.001483 1.006553e-07
## 3 0.01497079 3.659011 7.797879e-05 50 0.001483 2.564600e-05
## 4 0.65480456 4.069763 7.797879e-05 50 0.001483 3.809441e-05
## 5 0.67794988 2.674251 7.797879e-05 50 0.001483 3.868937e-07
## S04_Total_spkd S04_expctd_spkd spk_recovery S04_spks_flag
## 1 4.328042e-04 3.910426e-04 110.6796 YES
## 2 8.110539e-05 7.807945e-05 103.8755 YES
## 3 1.092581e-04 1.036248e-04 105.4362 YES
## 4 1.215231e-04 1.160732e-04 104.6952 YES
## 5 7.985315e-05 7.836568e-05 101.8981 YES
```

#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on

```
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)
```

spksbar



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_spks <- QAspks %>%
  group_by(S04_spks_flag) %>%
  summarise(no_rows = length(S04_spks_flag))
Perc_spks$Total <- length(QAspks$S04_spks_flag)
Perc_spks$Percent <- (Perc_spks$no_rows / Perc_spks$Total)*100
head(Perc_spks)
```

```
## # A tibble: 1 x 4
##   S04_spks_flag no_rows Total Percent
##   <chr>         <int> <int>   <dbl>
## 1 YES              5      5     100
```

Make final dataframe with IDs

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(sampledat2$Sample_ID), '_ ', fixed=TRUE)))
```

```
## Warning in rbind(c("100", "GWI", "202305", "WC", "LysB", "20cm"), c("101", :
## number of columns of result is not a multiple of vector length (arg 5)
```

```
colnames(IDs) <- c("Analysis_No" , "Site", "Date", "Zone", "Replicate", "Depth")
head(IDs)
```

```
##   Analysis_No Site   Date Zone Replicate Depth
## 1         100  GWI 202305  WC      LysB   20cm
## 2         101  GWI 202305  WC      LysB   45cm
## 3         102  GWI 202305  WC      LysC   10cm
## 4         103  GWI 202305  WC      LysC   20cm
## 5         104  GWI 202305  SW        A    104
## 6         105  GWI 202305  SW        B    105
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, sampledat2)
head(alldat)
```

```
##   Analysis_No Site   Date Zone Replicate Depth      Sample_ID
## 1         100  GWI 202305  WC      LysB   20cm 100_GWI_202305_WC_LysB_20cm
## 2         101  GWI 202305  WC      LysB   45cm 101_GWI_202305_WC_LysB_45cm
## 3         102  GWI 202305  WC      LysC   10cm 102_GWI_202305_WC_LysC_10cm
## 4         103  GWI 202305  WC      LysC   20cm 103_GWI_202305_WC_LysC_20cm
## 5         104  GWI 202305  SW        A    104   104_GWI_202305_SW_A
## 6         105  GWI 202305  SW        B    105   105_GWI_202305_SW_B
##   S04_ppm   Cl_ppm   S04_mM   Cl_mM salinity
## 1 1227.5264 9239.344 38.28841 260.6303 16.69552
## 2 1376.7938 8628.659 42.94429 243.4037 15.59201
## 3  958.8118 7841.280 29.90679 221.1927 14.16922
## 4 1411.9603 9624.554 44.04118 271.4966 17.39159
## 5 1574.3848 11468.676 49.10745 323.5170 20.72392
## 6 1609.8636 11757.423 50.21409 331.6621 21.24569
```

Make final dataframe with IDs

```
write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_PW_Processed_Cl_S04_202305_rerun_a.csv")
```

END